

Module 10 — Question Pool

OMSCS 6250 Computer Networks

Lesson 10: Internet Surveillance and Censorship

Identifying Censorship Techniques

Q1. [MCQ]

Which option BEST describes how DNS injection (poisoning), as used by the Great Firewall of China, blocks a censored domain?

- A. The censor withdraws the BGP route to the domain's authoritative name servers, so recursive resolvers worldwide can no longer reach them and every lookup for the domain times out.
- B. The censor drops every IP packet destined for the domain's server address at the AS border, taking down all services hosted on that address at once.
- C. An on-path censor watches DNS queries in flight and races a forged A-record back to the client, so the resolver accepts the poisoned answer before the legitimate authoritative response arrives; because it acts on the hostname, it can block one name without affecting other services sharing the same IP.
- D. An HTTP-aware proxy parses each request and returns a block page whenever the requested URL matches a banned list, giving per-page precision.

Correct answer: **C**

Why: DNS injection = on-path forged A-record that wins the race. The censor sees the query and returns a poisoned answer before the real authoritative response; being hostname-specific, it avoids the shared-IP collateral damage of IP blocking. BGP withdrawal, IP packet dropping, and proxy inspection are different techniques.

Q2. [MCQ]

Which option BEST describes how IP-based packet dropping censors a target, and its characteristic weakness?

- A. Routers inside the censoring network discard packets whose destination IP matches the blocked address, cutting off reachability to that host; because many domains can share one IP, this frequently overblocks co-hosted sites as collateral damage.
- B. The censor forges DNS answers so the client resolves the domain to a wrong address, while other names on the same shared IP stay reachable.
- C. The censor injects a TCP RST after inspecting the requested page, tearing down only the offending flow with single-object precision.
- D. The censor announces a more-specific BGP prefix for the target so global traffic is rerouted into a remote AS and discarded there.

Correct answer: **A**

Why: IP packet dropping = drop by destination IP => overblocking. Shared hosting/CDN means one IP serves many domains, so dropping it kills every co-hosted site. The other options describe DNS poisoning, TCP reset, and BGP manipulation.

Q3. [MCQ]

Which option BEST describes how TCP reset (RST) censorship works?

- A. The censor returns a forged A-record during name resolution, so the connection opens to a decoy server instead of the real destination.
- B. The censor withdraws the destination's BGP prefix, so the TCP handshake never completes because the SYN cannot be routed to the host.
- C. The censor drops all packets to the destination IP at the border router, so no TCP segment — including the initial SYN — ever reaches the host.
- D. An on-path device detects a connection to a forbidden destination and injects forged TCP RST packets toward one or both endpoints, tearing down the already-established flow so the transfer aborts mid-connection.

Correct answer: **D**

Why: TCP reset = inject forged RSTs to tear down an established flow. It acts on a connection that has already formed, unlike DNS poisoning (name resolution), BGP withdrawal (routing), or IP packet dropping (blocks the SYN itself).

Q4. [MCQ]

Which option BEST describes how proxy-based (HTTP-aware) content inspection censors traffic, and its main trade-off?

- A. The proxy blocks purely by destination IP address, which scales well across a country but cannot distinguish individual pages hosted on the same server.
- B. Traffic is routed through a proxy that parses each HTTP request and decides per-URL whether to allow or block it, giving the FINEST granularity — even a single page or object can be blocked — at the cost of poor scalability, since every flow must traverse the proxy.
- C. The proxy injects forged DNS responses so banned hostnames resolve to a block page, operating entirely at the name-resolution layer.
- D. The proxy withdraws BGP routes for banned content providers, making the block globally observable but very coarse.

Correct answer: **B**

Why: Proxy content inspection = most precise (per-URL/object), poorest scaling. The proxy parses HTTP per request, so it can block one page — but every flow must pass through it, which is expensive at country scale. The distractors describe IP blocking, DNS poisoning, and BGP manipulation.

Q5. [MCQ]

Which option BEST describes how BGP-based connectivity disruption works, and why it is comparatively EASY to detect?

- A. The censor withdraws or re-advertises BGP prefixes for the target networks, changing global control plane routing state; because these routing changes propagate to collectors such as RouteViews and RIPE RIS, they are visible from many vantage points worldwide.
- B. The censor drops packets at in-country routers while leaving the prefixes announced, so the disruption is visible only through active data-plane probes.
- C. The censor injects forged DNS answers at recursive resolvers, which is easy to detect because every resolver returns an identical poisoned record.
- D. The censor routes traffic through an HTTP proxy, which is easy to detect because the proxy returns a visible block page to each client.

Correct answer: **A**

Why: BGP disruption = withdraw/re-advertise prefixes => globally observable control-plane change. RouteViews/RIPE RIS feeds see the routing change from many vantage points. Option B is the data-plane technique (hard to see from outside); C and D mischaracterize DNS and proxy detection.

Q6. [MCQ]

Which option BEST describes how in-country (data-plane) packet filtering disrupts connectivity, and why it is HARDER to observe from outside the country?

- A. The censor withdraws BGP prefixes for the target, a change that appears immediately in global routing feeds observed from outside the country.
- B. The censor poisons DNS so queries resolve incorrectly, which any external resolver can confirm with a single lookup.
- C. Routers and firewalls inside the censoring country drop packets matching the target while the country's BGP prefixes stay announced globally; because the control plane looks unchanged, external observers must send active reachability probes (traceroute, ICMP, TCP) to detect it.
- D. The censor reroutes all national traffic through a proxy that advertises itself in BGP, making the filtering trivially observable worldwide.

Correct answer: **C**

Why: In-country filtering = data-plane drop with prefixes still announced => invisible in BGP alone. Since the control plane is unchanged, you need active reachability probing to detect it (this is why Augur-style measurement matters). Option A is the control-plane BGP technique; B and D mischaracterize DNS and proxy behavior.

Q7. [MCQ]

Which censorship technique is described as the MOST PRECISE — capable of blocking a single web page or even a specific object on a page — at the cost of POOR scalability?

- A. Immediate TCP reset of every incoming SYN, killing all flows rather than one page object.
- B. Packet dropping at the IP layer.

- C. DNS poisoning at the AS border.
- D. Proxy-based content inspection (HTTP-aware proxies that inspect every request).

Correct answer: **D**

Why: Proxy content inspection = surgical (block a page or object), poor scale. The proxy parses HTTP and decides per request, allowing per-URL granularity — but every flow has to traverse the proxy, which is expensive at country scale.

Why is DNS Manipulation Difficult to Measure?

Q8. [MCQ]

Identifying an INCONSISTENT or ANOMALOUS DNS response is, by itself, not enough to conclude that DNS manipulation is occurring. Why?

- A. DNS responses are encrypted by default and cannot be inspected.
- B. Legitimate causes (CDNs, geo-DNS, load-balancers, virtual hosting) produce different A-records to different resolvers — distinguishing benign inconsistency from intentional manipulation requires more signals.
- C. DNS manipulation typically returns identical answers to resolvers, so cross-resolver inconsistency is proof of benign behavior and anomalous A-records can safely be ignored during measurement.
- D. Only DNSSEC-signed responses can be classified; everything else is undecidable.

Correct answer: **B**

Why: Benign causes (CDN, geo-DNS, LB, vhosts) produce DNS inconsistency too. To call something manipulation, you need to rule out routine CDN/load-balancer behaviour — multiple signals beyond a single anomalous answer are required.

Q9. [TF]

Confirming DNS manipulation typically requires combining MULTIPLE signals (CONSISTENCY across resolvers + INDEPENDENT VERIFIABILITY of the answers) to rule out benign causes of response variation.

- True
- False

Correct answer: **True**

Why: Confirm DNS manipulation = consistency + independent verifiability. Cross-resolver consistency catches censor-injected answers; independent verifiability (e.g., TLS certs) confirms the IP actually serves the requested service.

Example Censorship Detection Systems and Their Limitations

Q10. [MCQ]

Why are home routers and consumer devices generally NOT suitable as measurement vantage points for global DNS censorship studies?

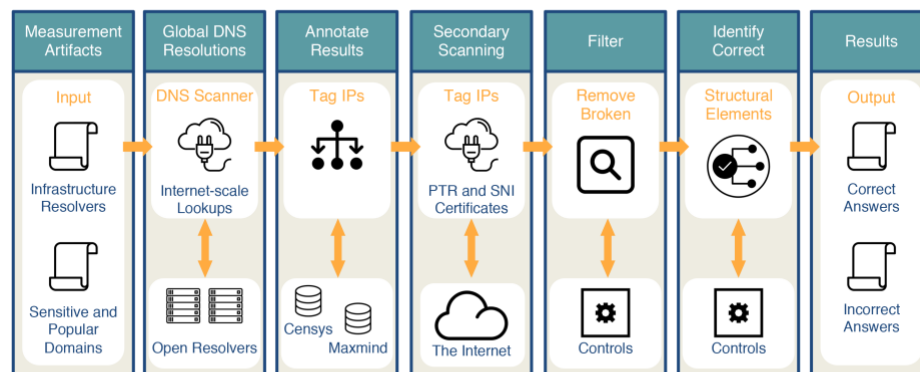
- A. They do not support TLS, so they cannot encrypt the measurement traffic.
- B. Their CPUs are too slow to perform DNS lookups at the required rate.
- C. Many such devices are open due to misconfiguration; involving them risks legal/personal danger for residents and produces biased measurements against the population most exposed to censorship.
- D. Home routers typically run inside the same AS as the censor, so DNS queries they issue are intercepted before leaving the local network, making censorship measurement from them structurally unreliable.

Correct answer: C

Why: Home routers = ethical risk + biased sample. Probing through residential devices can put residents at legal risk and skews measurement toward whoever happens to leave a misconfigured router open.

DNS Censorship: An Example Global Measurement Methodology

Q11. [MCQ]



Overview of DNS Resolution, Filtering, and Classification Georgia Tech

Reference Figure: Iris pipeline — resolve, annotate, filter, classify (Module 10)

In the Iris pipeline figure, why does the system perform additional PTR and TLS-with-SNI (Server Name Indication) scanning of returned IPs?

- A. Because TLS is required to query the Censys dataset.
- B. To verify that the recursive resolver is geographically inside the target country.
- C. To re-poison the DNS cache for cross-validation against the censor, forcing the resolver to replay injected answers so Iris can confirm poisoning by comparing the fetched certificates on each subsequent lookup.
- D. To avoid mislabeling legitimate virtual hosting as DNS inconsistency — Censys-fetched certs may differ from SNI-fetched certs for benign reasons, and the additional scans disambiguate those.

Correct answer: **D**

Why: PTR/SNI scans disambiguate virtual hosting from manipulation. Different certs across resolvers can be benign vhosting; SNI-aware fetches verify whether the returned IP actually serves the queried name.

Q12. [MCQ]

Iris restricts its set of DNS resolvers to INFRASTRUCTURE resolvers (ISP/cloud), not home routers. Why?

- A. Infrastructure resolvers are the only ones that respond to Iris's special test queries, because home routers silently discard any lookup not originating from their own local network.
- B. Infrastructure resolvers are intentionally part of the Internet's public infrastructure, minimizing ethical risk to citizens and producing more representative measurements.
- C. Home routers typically reject queries from outside their LAN, so they cannot be used.
- D. Iris cannot decrypt responses from home routers.

Correct answer: **B**

Why: Infrastructure resolvers (ISP/cloud) = ethical + representative. They're intentionally public DNS infrastructure, so measuring through them avoids exposing individuals and yields more representative country-scale views.

Q13. [MCQ]

To decide whether a resolver's DNS answer for a domain is correct or manipulated, Iris evaluates the returned IP against two families of signals. Which option correctly identifies the signals in each family, the role each family plays, and how Iris combines them?

- A. Consistency signals — IP address, Autonomous System, HTTP content, HTTPS certificate, and PTR records for CDNs — test whether the returned IP's network/infrastructure/content properties match high-confidence baselines seen from other vantage points; independent-verifiability signals — the HTTPS certificate fetched with and without SNI (Server Name Indication) — use external data to confirm the returned IP itself serves a valid, browser-trusted certificate for the queried domain. A response is accepted as correct if any single consistency or verifiability metric is satisfied; only if none is satisfied is it flagged as manipulated.
- B. “Consistency” and “independent verifiability” are two names for the same cross-resolver agreement test; a response is labeled manipulated only when every resolver returns a different IP address.
- C. Independent-verifiability signals are the AS number and IP prefix (confirming the IP is on the right network), while consistency signals are the HTTPS certificates; a response is correct only if both families are fully satisfied.
- D. Iris relies on a single signal — DNSSEC validation of the returned records — and treats any domain that is not DNSSEC-signed as manipulated.

Correct answer: **A**

Why: Iris labels each answer using two metric families (Global Measurement of DNS Manipulation). Consistency metrics (IP address, AS, HTTP content, HTTPS certificate, PTRs for CDNs) compare the returned IP to high-confidence baselines and across vantage points

— but benign CDNs, geo-DNS, load-balancing, and virtual hosting create legitimate inconsistency, so consistency alone over-flags. Independent-verifiability metrics (HTTPS certificate with and without SNI) use external data to confirm the returned IP itself serves a valid, browser-trusted cert for the queried name, catching those false positives. The rule is permissive: satisfying any one metric (either family) marks the response correct; only when none passes is it manipulated. B conflates the two families and invents an all-resolvers-differ rule; C swaps which signals belong to which family and wrongly requires both; D reduces Iris to DNSSEC-only, which it is not.

Q14. [MCQ]

Which approach to connectivity disruption is described as EASIER to detect because it modifies the global control-plane state of the network?

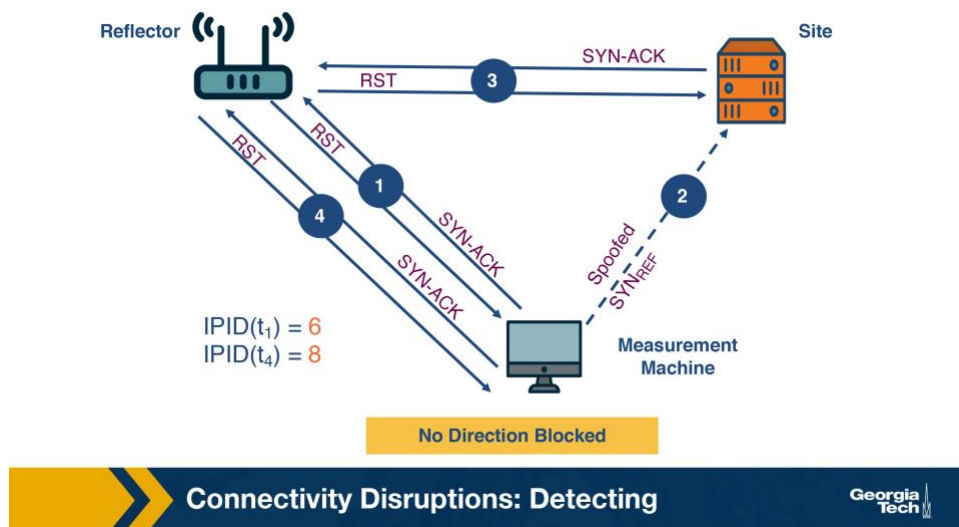
- A. Deep packet inspection inside ISPs at the application layer.
- B. Packet filtering at routers and firewalls inside the censoring country.
- C. Routing disruption via BGP withdrawals or re-advertisements — visible to many vantage points globally.
- D. DNS-based blocking using injected A records to local clients.

Correct answer: **C**

Why: BGP-based disruption = globally observable from many vantage points. Routing changes show up in RouteViews/RIPE RIS feeds worldwide; in-country packet filtering is invisible from outside the country.

Connectivity Disruptions: Detection

Q15. [MCQ]



Reference Figure from Canvas Module: Augur — IP-ID side channel, no-filtering case, IP-ID increments by 2 (Module 10)

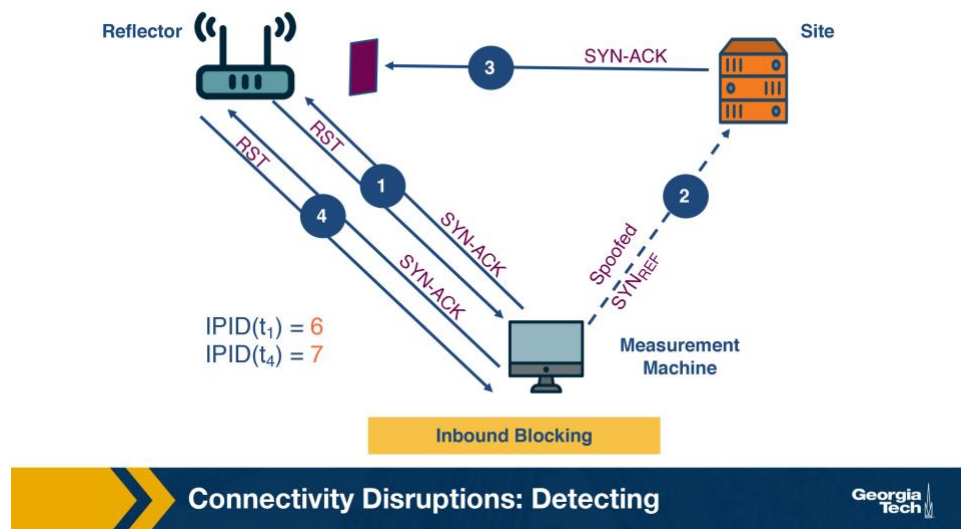
In Augur's no-filtering scenario, the reflector's IP-ID rises by TWO across the measurement window. Why?

- A. Because TCP requires two RSTs per connection close.
- B. Because the reflector doubles its IP-ID counter on each probe it answers, independent of whether any spoofed traffic reached it in that window.
- C. The probe itself triggers one RST from the reflector (+1) and the spoofed perturbation triggers another reflector RST to the site (+1), totaling +2.
- D. Because the measurement machine sends two SYN-ACKs at t4 to be safe.

Correct answer: **C**

Why: No filtering => +2: one RST from probe to reflector + one from site to reflector. Augur's IP-ID side channel: the probe triggers a RST (+1), and the spoofed perturbation triggers another RST from the site (+1) — total +2.

Q16. [MCQ]



Reference Figure from Canvas Module: Augur — inbound blocking, IP-ID +1 only (Module 10)

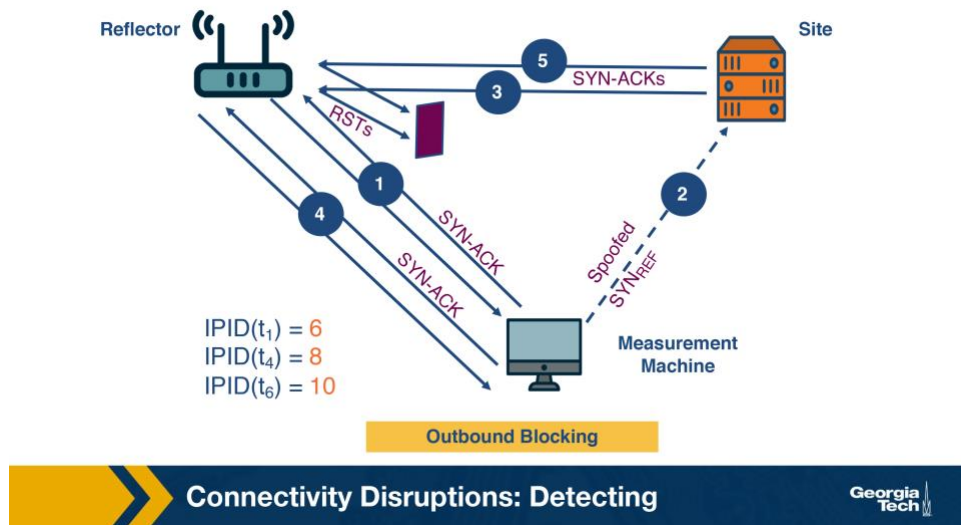
What does Augur observe when INBOUND filtering exists on the path from the site to the reflector?

- A. The reflector stops responding to all probes for the duration of the test.
- B. IP-ID at t4 increases by TWO — identical to the no-filtering case.
- C. IP-ID wraps to zero because the reflector resets its counter.
- D. IP-ID at t4 increases by only one — the site's SYN-ACK never reaches the reflector, so no second RST is generated.

Correct answer: **D**

Why: Inbound block => site's SYN-ACK never reaches reflector => only +1. With inbound filtering, the spoofed SYN-ACK from the site is dropped on the way to the reflector, so no second RST is generated.

Q17. [MCQ]



Reference Figure from Canvas Module: Augur — outbound blocking, retransmissions inflate IP-ID (Module 10)

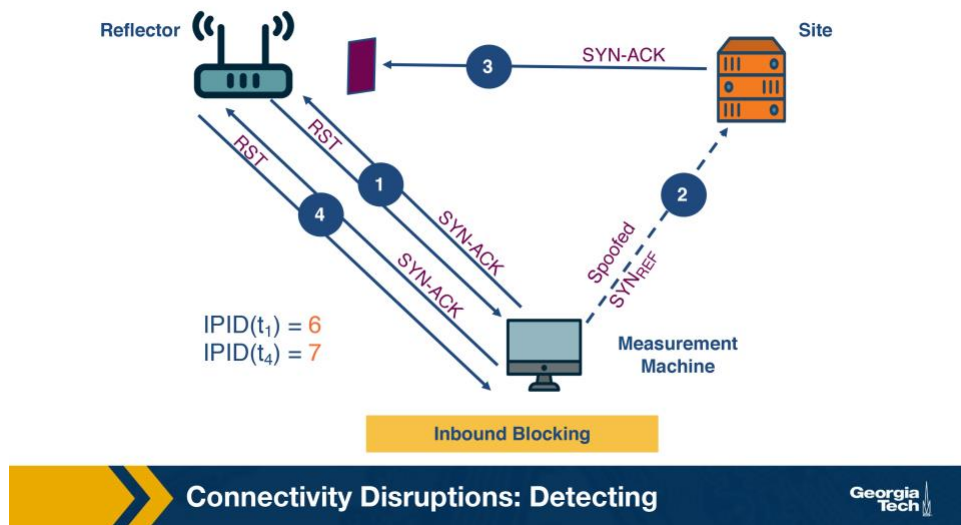
How does Augur DISTINGUISH outbound blocking (filtering on the path from reflector toward the site) from the no-filtering case?

- A. The site never receives the reflector's RST and retransmits SYN-ACKs; this causes additional increments in the reflector's IP-ID beyond the baseline.
- B. Outbound blocking causes the reflector's IP-ID to decrement, because the dropped SYN-ACKs force its global counter to roll backward below the measured baseline.
- C. The measurement machine sees a duplicate RST with the same IP-ID.
- D. Outbound blocking is indistinguishable from no filtering at the IP-ID layer.

Correct answer: **A**

Why: Outbound block => retransmits inflate IP-ID beyond baseline. The site doesn't receive the reflector's RST, so it retransmits its SYN-ACKs; each retransmit triggers another RST from the reflector, increasing IP-ID by more than 2.

Q18. [MCQ]



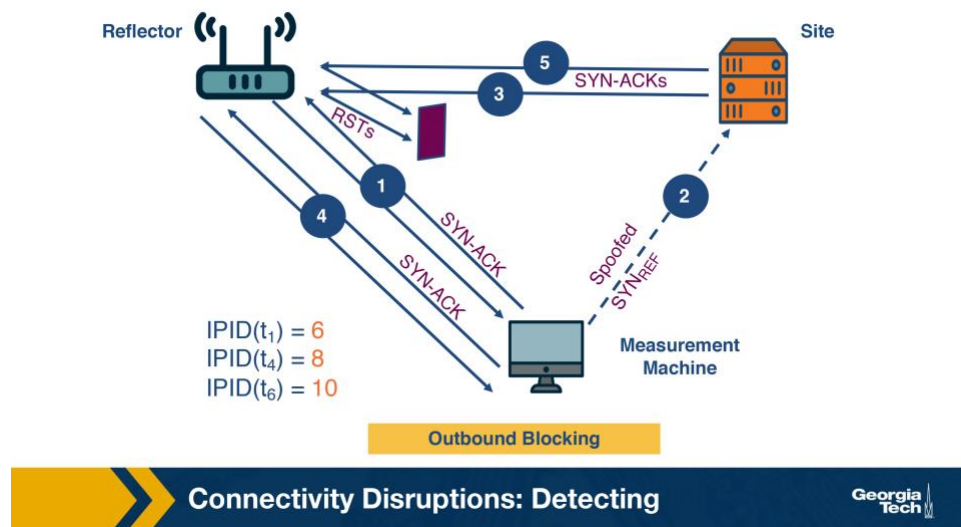
Scenario: Augur probes a reflector at t_1 and gets IP-ID = 42. After perturbation, t_4 gives IP-ID = 43. What does Augur most likely infer?

- A. No filtering — the IP-ID increment should be exactly 2 without filtering, since the probe and the site's SYN-ACK each advance the counter once.
- B. Inbound blocking on the path from site to reflector — the site's SYN-ACK never reached the reflector, so only the probe-triggered RST counted.
- C. Outbound blocking — only outbound blocking can produce +1.
- D. The reflector is no longer using a global IP-ID counter.

Correct answer: **B**

Why: +1 increment => inbound blocking. Only one of the two expected RSTs reached the reflector; the path from site to reflector dropped the SYN-ACK, so the second RST never fired.

Q19. [MCQ]



Reference Figure from Canvas Module: Augur — interpreting IP-ID jump of 4 (Module 10)

Scenario: Augur probes a reflector at t_1 and gets IP-ID = 100; at t_4 it observes IP-ID = 104 (jump of 4). Which interpretation aligns with the lesson?

- A. No filtering exists; the extra increments are random noise from unrelated traffic hitting the reflector, so a jump of 4 carries no information about path blocking.
- B. The reflector lost its global IP-ID counter.
- C. Outbound blocking — the site retransmits SYN-ACKs because it never received the reflector's RST, inflating the IP-ID beyond the no-filtering baseline.
- D. Inbound blocking with delayed responses.

Correct answer: **C**

Why: +4 jump => outbound blocking + retransmits. Multiple SYN-ACK retransmits from the site each triggered a RST at the reflector; the inflated count is Augur's signature for outbound filtering.

Q20. [TF]

Augur's IP-ID side-channel technique assumes a SINGLE GLOBAL IP-ID counter at the reflector — measurements may be misleading on reflectors that use per-flow IDs or that experience heavy cross-traffic.

- True
- False

*Correct answer: **True***

Why: Augur assumes a SINGLE GLOBAL IP-ID counter. Modern OSes increasingly use per-flow or randomized IDs, breaking the technique on those reflectors; cross-traffic at the reflector also adds noise.

Applied Scenarios: Designing Censorship Detection

Q21. [MCQ]

Scenario — Suspected DNS manipulation. A regional ISP is suspected of injecting fake DNS answers for a set of news domains, but it claims the differing answers are just normal CDN and load-balancer behavior. You control infrastructure (ISP/cloud) resolvers inside and outside the region. Which measurement approach would BEST confirm manipulation while ruling out benign response variation?

- A. Query the domains through many geographically diverse infrastructure resolvers and collect BOTH consistency signals (AS number, IP prefix, HTTP content) AND independent-verifiability signals (TLS certificates fetched with and without SNI [Server Name Indication], plus PTR records) to check whether the returned IP actually serves the queried name.
- B. Use a geographically diverse set of infrastructure (ISP/cloud) resolvers and collect the consistency signals across them — matching AS numbers, IP prefixes, and HTTP body content — then flag any domain on which the resolvers disagree about the returned A-record as manipulated.
- C. Query the domains through many infrastructure resolvers and, for each returned IP, open a TLS connection and inspect the served certificate; treat any domain whose fetched certificate fails to validate for the requested hostname as evidence of injection.
- D. Restrict the study to DNSSEC-signed domains, validate the RRSIG chain returned by each resolver inside and outside the region, and classify a domain as censored whenever its signature fails to validate — treating unsigned domains as inconclusive.

*Correct answer: **A***

Why: Confirm manipulation with consistency PLUS independent verifiability. A single differing A-record is exactly what benign CDNs, geo-DNS, load-balancers, and virtual hosting produce, so you need both cross-resolver consistency signals AND an independent check that the returned IP truly serves the queried name (TLS certs with/without SNI, PTR) — this is what A does. B uses consistency

only: it flags any resolver disagreement as manipulation, misclassifying benign CDN/geo-DNS variation as censorship. C uses the certificate check only: with no consistency baseline, benign hosts whose certs legitimately fail to match get over-flagged. D restricts the study to DNSSEC-signed domains, so it is blind to the overwhelming majority of unsigned domains and mostly returns 'inconclusive.'

Q22. [MCQ]

Scenario — Suspected nationwide connectivity disruption. During civil unrest a government is suspected of ordering a large-scale Internet shutdown. You are located entirely OUTSIDE that country. Which data-collection strategy lets you both detect the disruption AND distinguish a control-plane mechanism (BGP route withdrawals) from a data-plane mechanism (in-country packet filtering)?

- A. Collect BGP updates from globally distributed collectors (RouteViews, RIPE RIS) to catch prefix withdrawals, AND run active data-plane probes (traceroute, ICMP echo, TCP connection attempts) to catch filtering that leaves routes announced; correlate the timing of the two.
- B. Continuously monitor BGP updates from RouteViews and RIPE RIS for prefix withdrawals and AS-path changes; as long as the country's prefixes stay announced throughout the event window, conclude that connectivity is intact and that no shutdown of any kind occurred.
- C. Run active data-plane probes only — traceroute, ICMP echo, and TCP connection attempts — from many external vantage points; because a routing withdrawal ultimately shows up as unreachable hops, the probe results by themselves reveal both the outage and whether it was caused by BGP changes or in-country filtering.
- D. Track DNS resolution latency and timeout rates against in-country recursive resolvers over time; a coordinated shutdown appears as a sharp spike in query failures, and the shape of that failure pattern distinguishes a control-plane route withdrawal from data-plane packet filtering.

Correct answer: **A**

Why: Combine control-plane and data-plane views. BGP feeds (RouteViews/RIPE RIS) catch prefix withdrawals; active probes (traceroute/ICMP/TCP) catch filtering that leaves routes announced; correlating their timing tells the mechanisms apart — that is A. B monitors BGP only, but in-country packet filtering leaves prefixes announced, so 'prefixes still up' does NOT mean connectivity is intact — it misses data-plane blocks entirely. C probes the data plane only: it can see hosts are unreachable but cannot, by itself, separate a control-plane route withdrawal from in-country filtering. D watches DNS failure rates, which neither prove a nationwide outage nor distinguish a route withdrawal from packet filtering — the 'failure shape' does not encode the mechanism.

Q23. [MCQ]

Scenario — Remote directional filtering with no local vantage point. You want to determine whether a remote network is blocking traffic to a particular site, AND in which direction (inbound toward the site vs. outbound from the site), but you have NO measurement machine inside either the client's or the site's network. Which technique — and which observable — would you design for this?

- A. Augur's IP-ID side channel: pick a reflector with a single global IP-ID counter, record a baseline IP-ID, send a spoofed SYN-ACK carrying the site's address, then re-probe — a +2 increment means no filtering, +1 means inbound blocking, and an inflated jump (>2) means outbound blocking.

- B. Run repeated traceroutes from several external vantage points toward the site and infer direction from where the probes die: if the trace terminates before the site's border network the block is inbound, and if the returning TTL-exceeded messages go missing the block is outbound.
- C. Use the IP-ID side channel but read direction from RST timing instead of counter values — spoof a SYN to the reflector as the site and measure the round-trip to the RST: a fast RST means no filtering, a delayed RST means inbound blocking, and a lost RST means outbound blocking.
- D. Pick a reflector with a single global IP-ID counter and send a spoofed SYN-ACK bearing the site's address, but map the increments the other way: a +1 advance indicates outbound blocking (only the reflector's own RST moved the counter) while an inflated jump indicates inbound retransmission by the site.

Correct answer: **A**

Why: Augur's IP-ID side channel infers direction remotely from the size of the reflector's IP-ID advance: +2 = no filtering, +1 = inbound blocking (the site's SYN-ACK never reached the reflector), inflated (>2) = outbound blocking (the site retransmits SYN-ACKs, each triggering another RST) — that is A. B uses traceroute, which gives reachability but not direction: where a probe dies does not reveal inbound vs outbound filtering. C keeps the side channel but reads RST round-trip timing instead of the IP-ID counter, and RST latency does not encode blocking direction, so the inference is invalid. D uses the right technique but inverts the mapping — calling +1 'outbound' and an inflated jump 'inbound,' exactly backwards from the actual counter behavior.

Applied Scenarios: Thinking Like a Censor

Q24. [MCQ]

Scenario — Surgical page-level blocking. A censor operating a national gateway wants to block ONE specific article on a large news site while keeping the rest of that site — and every other site sharing its CDN IP address — fully reachable. The censor's gateway can terminate and inspect TLS. Which single technique BEST satisfies this goal of maximum granularity with minimal collateral damage?

- A. Announce a more-specific BGP route for the site's server prefix and null-route it, dropping all packets to that address.
- B. Poison DNS for the site's domain so the hostname resolves to a block page.
- C. Inject a TCP RST into every connection opened to the site's server IP.
- D. Pass the traffic through an intercepting content-inspection proxy that parses each request and returns a block page only for the target article's URL, forwarding everything else untouched.

Correct answer: **D**

Why: Granularity ranking. IP/BGP null-routing (A) and RST injection (C) act on an address and take down the whole host and every CDN-colocated site; DNS poisoning (B) blocks the entire domain. Only a content-inspecting proxy (D) reaches URL/object granularity, so it alone blocks one page while leaving the rest of the site and its IP-neighbors reachable — at the cost of the most expensive, highest-latency infrastructure.

Q25. [MCQ]

Scenario — Deniable degradation during unrest. A government wants people to give up on a video platform because it is too slow to be usable, but wants to AVOID an obvious hard block: the disruption

should look like ordinary network congestion, be hard to attribute, and not trip circumvention tools or censorship-measurement alarms that fire on outright blocking. Which approach BEST satisfies this goal?

- A. Throttle: rate-limit / shape bandwidth to the platform's IP ranges and Server Name Indication (SNI) so connections still complete but media stalls, mimicking congestion.
- B. Withdraw the BGP routes covering the platform's prefixes nationwide.
- C. Return NXDOMAIN for the platform's domain at all in-country recursive resolvers.
- D. Inject TCP RSTs on every connection to the platform to tear down sessions.

Correct answer: **A**

Why: The goal is plausible deniability, not a clean block. Throttling (A) leaves the service nominally “up” — pages and TCP handshakes succeed — so the failure is indistinguishable from congestion and is not flagged by tools that detect reachability loss. BGP withdrawal (B), DNS NXDOMAIN (C), and RST injection (D) all produce a crisp, easily measured, easily attributed hard failure — and DNS/RST blocks are trivially circumvented (alternate resolver, ignore RSTs), defeating the “make it quietly unusable” goal.

Q26. [MCQ]

Scenario — Cheap, central, high-churn blocklist. A censor must add hundreds of newly-registered domains to a national blocklist every day, wants the cheapest mechanism to operate and update from a central point without reconfiguring every ISP's core routers, and explicitly ACCEPTS that technically sophisticated users will bypass it. Which technique BEST fits these goals?

- A. DNS manipulation at the country's recursive resolvers: push each new domain to a central blocklist that returns NXDOMAIN or a block-page IP.
- B. Deploy intercepting content-inspection proxies that MITM TLS and match each blocked domain per request.
- C. BGP null-route the IP prefixes hosting each blocked domain.
- D. Deploy DPI boxes that RST connections whenever a blocked domain appears in the TLS Server Name Indication (SNI).

Correct answer: **A**

Why: Match the tool to the goal set: cheap, centrally managed, high-churn, circumvention-tolerated. DNS blocking (A) is the low-cost, domain-granular, centrally-updatable mechanism — a blocklist edit propagates instantly and needs no router changes — and its well-known weakness (users switch resolvers or use DoH) is explicitly acceptable here. Proxies (B) and SNI-DPI (D) are far costlier per-flow inspection infrastructure, overkill when circumvention is tolerated; BGP null-routing (C) is address- not domain-granular, causes CDN collateral, breaks as hosting IPs rotate, and touches routing infrastructure the censor wanted to leave alone.